1.11 Quadratic Modelling

Worked Solutions

1 a)

$h = 19.6 \times 2 - 4.9 \times 4 = 19.6$ m

1 b)

$14.7 = 19.6t - 4.9t^2$

$4.9t^2 - 19.6t + 14.7 = 0$

$t = \frac{19.6 \pm \sqrt{384.16 - 288.12}}{9.8}$ so $t = 1\text{s}, t = 3\text{s}$

1 c)

$t = 1$ is when it reaches a height 14.7m on the way up and $t = 3$ is when it drops back to the height of 14.7m on the way back down.

2 a)

$0 = 2t(5 - t)$ so $t = 5\text{s}$

2 b)

$-2t^2 + 10t = -2(t^2 - 5t)$

$= -2[(t - \frac{5}{2})^2 - (-\frac{5}{2})^2] = -2(t - \frac{5}{2})^2 + \frac{25}{2}$

2 c)

Max at $(\frac{5}{2}, \frac{25}{2})$ so max height is 12.5m after 2.5s has passed.

2 d)

$-2t^2 + 10t + 1$

2 e)

$(\frac{5}{2}, \frac{27}{2})$

3 a)

The posts are 0.5m high

3 b)

$\frac{1}{100}x^2 - \frac{1}{40}x + \frac{1}{2} = \frac{1}{100}[x^2 - \frac{5}{2}x + 50]$

$= \frac{1}{100}[(x - \frac{5}{4})^2 - (-\frac{5}{4})^2 + 50]$

$= \frac{1}{100}[(x - \frac{5}{4})^2 + \frac{775}{16}] = \frac{1}{100}(x - \frac{5}{4})^2 + \frac{31}{64}$

3 c)

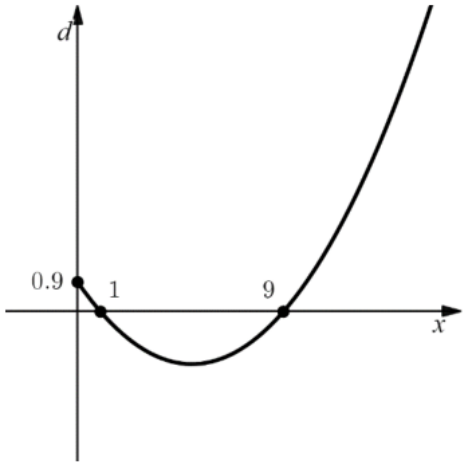
Horizontal position of the minimum height of the rope is 1.25m from the left post

Vertical position of the minimum height of the rope $\frac{31}{64}\text{m} \approx 0.48\text{m}$ (2 SF)

3 d)

The rope ends at the top of the right post which is 2.5m from the left post. So $x > 2.5$ would be modelling beyond the length of the rope.

4 a)



4 b)

$\frac{1}{2}(1+9) = 5$ when $x = 5, d = -1.6$

1.6 metres below the surface.

4 c)

Between $x = 1$ and $x = 9$ so 8 metres

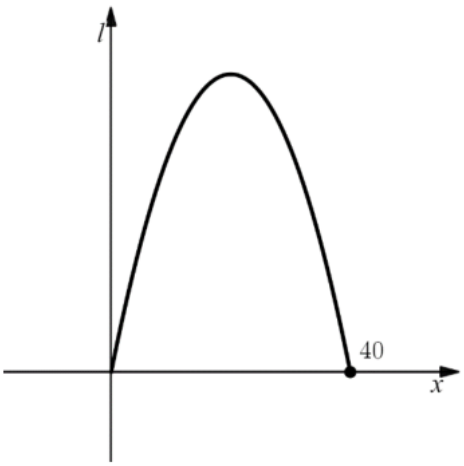
4 d)

1 m

4 e)

After 9 metres from the poolside, the model suggests that the swimmer is leaving the water, so $x > 9$

5 a)



5 b)

$\frac{40}{2} = 20 \Rightarrow y = 1000$ So $I = 1000 - \frac{5}{2}(x-20)^2$

5 c)

£1000

5 d)

£20

5 e)

$1000 - \frac{5}{2}(x-20)^2 = 600 \Rightarrow \frac{5}{2}(x-20)^2 = 400$

$\Rightarrow (x-20)^2 = 160 \Rightarrow x = 20 \pm \sqrt{160}$

so $x = £7.35$ or $x = £32.65$

6 a)

$y = \frac{15}{2}x - \frac{25}{2} - \frac{5}{8}x^2 = -\frac{5}{8}[x^2 - 12x + 20]$

$= -\frac{5}{8}[(x-6)^2 - 6^2 + 20]$

$= -\frac{5}{8}[(x-6)^2 - 16]$

$= -\frac{5}{8}(x-6)^2 + 10$ so $x = 6$

Best profit when charging £6 per book

6 b)

Max profit of £0.10 per book

6 c)

$-\frac{5}{8}[x^2 - 12x + 20] = 0$

$x^2 - 12x + 20 = 0$

$(x-2)(x-10) = 0$

No profit when charging £2 or £10 per book

6 d)

y-intercept: $y = -12.5$ so a 12.5p loss per book

7 a)

$-\frac{2}{45}x^2 + \frac{8}{3}x = -\frac{1}{10}x^2 + 3x + \frac{15}{2}$

$\frac{1}{18}x^2 - \frac{1}{3}x - \frac{15}{2} = 0$

$x = \frac{\frac{1}{3} \pm \sqrt{\frac{1}{9} + \frac{5}{3}}}{\frac{1}{9}} = \frac{\frac{1}{3} \pm \frac{4}{3}}{\frac{1}{9}}$ so $x = 15$ or $x = -9$

x can't be negative so at 15 days.

7 b)

$-\frac{1}{10}x^2 + 3x + \frac{15}{2} = -\frac{1}{10}[x^2 - 30x - 75]$

$= -\frac{1}{10}[(x-15)^2 - (-15)^2 - 75]$

$= -\frac{1}{10}[(x-15)^2 - 300] = -\frac{1}{10}(x-15)^2 + 30$

So maximum height is 30 cm

7 c)

The quadratics reach a maximum then reduce, whilst the plants will either keep growing or reach a fixed height.

8)

Max at $(2, 10)$

$h = k(t-2)^2 + 10$ for some constant k

At $(0,0) : 0 = k(-2)^2 + 10$ so $k = -2.5$

$h = -2.5(t-2)^2 + 10$

$h = -2.5(t^2 - 4t + 4) + 10$

$h = -2.5t^2 + 10t - 10 + 10$

$h = -2.5t^2 + 10t$

9 a)

At $(a, 0) : 0 = -\frac{1}{8}a^2 + \frac{11}{8}a - \frac{57}{32}$ so $a = 1.5$m or $a = 9.5$m. Since it is projected from a then $a = 1.5$m.

9 b)

$9.5 - 1.5 = 8$m

9 c)

Max occurs when $x = 5.5$,

so $y = -\frac{1}{8}(5.5)^2 + \frac{11}{8}(5.5) - \frac{57}{32} = 2$m above the ground

9 d)

At $x = 7.5$ the ball is at height

$y = -\frac{1}{8}(7.5)^2 + \frac{11}{8}(7.5) - \frac{57}{32} = 1.5$m which is higher than the tree so it doesn't hit it.

10)

Max point at $(175, 140)$

$y = k(x - 175)^2 + 140$ for some constant k

Left most arch start at $(0, 0)$:

$0 = k(-175)^2 + 140$ so $k = -\frac{4}{875}$

$y = -\frac{4}{875}(x - 175)^2 + 140$

$y = -\frac{4}{875}(x^2 - 350x + 30625) + 140$

$y = -\frac{4}{875}x^2 + \frac{8}{5}x = \frac{4}{875}x(350 - x)$